Link - https://rj42-coder.github.io/e-waste-epr-calculator/E-Waste (Management) Rules, 2022: Compliance Practices of Leading Indian Electronics Manufacturers

A Review of Extended Producer Responsibility Practices Among Select Indian Electronics Manufacturers

Student: *Riya Jamsutkar (Roll No: 24102A0053)* | ***Course:*** *Electronic Waste Management & Circular Economy*

Prepared with reference to the CPCB FAQ on the E-Waste (Management) Rules, 2022, and amendments thereafter (23 January 2024)

Abstract

The E-Waste (Management) Rules, 2022, in force in India since 1 April 2023, place the core burden of electronic-waste management on Manufacturers, Producers, Refurbishers, and Recyclers through a system of Extended Producer Responsibility (EPR) administered via the Central Pollution Control Board's (CPCB) E-Waste EPR Portal. This paper reviews the compliance posture of five prominent electronics manufacturers operating in India — Dixon Technologies, Syrma SGS Technology, VVDN Technologies, Tata Electronics, and Foxconn (Hon Hai) India — with reference to their publicly available Business Responsibility and Sustainability Reports (BRSR), company e-waste policies, and regulatory filings. The review finds that all five companies formally recognise EPR applicability, but that the depth of public disclosure varies considerably between SEBI-listed entities and unlisted or foreign-parented subsidiaries. The paper concludes that documentary linkage with named, state-authorised recyclers offers the clearest compliance trail under the Rules, and that BRSR-linked disclosure is progressively narrowing the transparency gap for unlisted entities.

Keywords: E-Waste (Management) Rules 2022, Extended Producer Responsibility, CPCB, Compliance, Electronics Manufacturing, India

1. INTRODUCTION AND REGULATORY BACKGROUND

The E-Waste (Management) Rules, 2022, notified under the Environment (Protection) Act, 1986 and in force since 1 April 2023, govern the manufacture, sale, purchase, refurbishing, dismantling, recycling, and processing of electrical and electronic equipment (EEE) in India. The Rules cover 106 categories of EEE across seven equipment classes and place the core compliance burden on Manufacturers, Producers, Refurbishers, and Recyclers, each of whom must register on the CPCB's E-Waste EPR Portal.

The central compliance mechanism is Extended Producer Responsibility (EPR): producers must meet phase-wise e-waste recycling targets — 60% of generated waste for FY 2023-24 and FY 2024-25, rising to 70% by FY 2026-27 and 80% from FY 2027-28 onward — by sourcing EPR certificates exclusively from CPCB-registered recyclers. The Rules also mandate RoHS compliance (restricting Lead, Mercury, Cadmium, Hexavalent Chromium, and brominated flame retardants), quarterly and annual returns, and audits of stakeholders.

2. MATERIAL AND METHOD

This review examines five prominent electronics manufacturers operating in India against the obligations set out above. The assessment draws on publicly available Business Responsibility and Sustainability Reports (BRSR), company-published e-waste management policies, and regulatory filings. Companies were selected to represent a cross-section of the sector: SEBI-listed electronics manufacturing services (EMS) providers, an India-based product engineering firm, a large unlisted

domestic-group subsidiary, and an Indian operation of a global contract manufacturer.

3. COMPANY PROFILES

3.1 Dixon Technologies (India) Ltd.

India's largest electronics manufacturing services (EMS) provider — LED TVs, washing machines, mobile phones, and lighting products.

EPR applicability: Confirmed applicable to the company's operations.

• Acknowledges EPR applicability and states its waste collection plan is aligned with the EPR plan filed with State Pollution Control Boards.

• All manufacturing facilities certified Zero Waste to Landfill (ZWL); waste is handed only to CPCB/SPCB-approved disposal agencies.

• Discloses EPR and waste data annually through its SEBI-mandated Business Responsibility and Sustainability Report (BRSR).

Source(s): Dixon Technologies BRSR, FY 2023-24 (filed with NSE, 3 September 2024).

3.2 Syrma SGS Technology Ltd.

Listed EMS and design-led manufacturing company serving automotive, industrial, medical, and consumer electronics segments.

EPR applicability: Confirmed — registered as Producer.

• Formally answers "Yes" to EPR applicability in its BRSR and confirms its waste collection plan is in line with the EPR plan submitted to Pollution Control Boards.

• Has a signed agreement with Global Waste Solution, an HSPCB-authorised dismantler and recycler, to handle collection, transportation, storage, segregation, dismantling, and recycling of e-waste.

• Publishes a dedicated EPR Compliance page listing state-wise authorised collection points and explicitly instructs customers not to hand e-waste to informal ("kabadi") channels.

• FY 2025-26 BRSR reports continued third-party assurance of its environmental and waste-management disclosures.

Source(s): syrmasgs.com/epr-compliance; Syrma SGS BRSR FY 2023-24 and FY 2025-26 filings (NSE).

3.3 VVDN Technologies

India-based product engineering and electronics manufacturing company (networking, IoT, camera, and telecom hardware).

EPR applicability: Confirmed — EPR Plan filed.

• Publishes an E-Waste Management policy committing to full compliance with the E-Waste (Management) Rules and their amendments.

• Partnered with M/s E-Waste Recyclers India, a recycler/dismantler registered with the Haryana State Pollution Control Board, for pan-India collection and recycling/dismantling of e-waste.

• Collection and channelisation of e-waste is handled entirely through this registered-recycler partnership rather than informal scrap channels.

Source(s): vvdntech.com/e-waste-management.

3.4 Tata Electronics Pvt. Ltd.

Tata Group subsidiary manufacturing semiconductors and electronic components/assemblies (including smartphone assembly) in India.

EPR applicability: Governed by the Tata Group E-Waste Management Policy.

• As a Tata group company, operates under the group-wide E-Waste Management Policy Commitment, which requires all Tata operating companies to obtain EPR registration and comply with the E-Waste Rules, 2022-2024.

• The group policy also mandates employee sensitisation on e-waste hazards and safe handling.

• Being an unlisted subsidiary, Tata Electronics does not file a separate public BRSR; its compliance is disclosed at the group-policy level rather than through company-specific waste data.

Source(s): Tata group 'E-Waste Management Policy Commitment Document' (tatapower.com/who-we-are/company-documents).

3.5 Foxconn India (Hon Hai group)

Contract electronics manufacturer operating facilities in Tamil Nadu, Karnataka, and Telangana, producing mobile phones and electronic components for global brands.

EPR applicability: Applicable as Manufacturer/Producer under Schedule I.

• As a manufacturer of Schedule I equipment (mobile phones, components) in India, is required to register as Manufacturer/Producer on the CPCB EPR Portal and meet phase-wise recycling targets.

• Follows its parent Hon Hai group's global EHS and sustainability framework, which requires engagement of authorised recyclers for electronic and process waste.

• India-specific EPR filings and waste data are not separately published; disclosure is largely through the Hon Hai group's global sustainability reporting rather than an India BRSR.

Source(s): General industry and CPCB EPR framework references; Hon Hai/Foxconn group sustainability disclosures.

4. COMPARATIVE SUMMARY

Table 1 consolidates the EPR registration status, recycler partnerships, and public disclosure posture of the five companies reviewed.

Table 1

EPR Registration, Recycler Partnerships, and Disclosure Posture of Reviewed Companies

Company

EPR Registration

Registered Recycler Partner(s)

Public Disclosure

Dixon Technologies (India) Ltd.

Yes — Producer/Manufacturer

Disposal via CPCB/SPCB-approved agencies; Zero Waste to Landfill at all plants

BRSR (FY 2023-24) filed with NSE/BSE

Syrma SGS Technology Ltd.

Yes — Producer

M/s Global Waste Solution (HSPCB-authorised dismantler/recycler)

BRSR filed annually; EPR page on company website

VVDN Technologies

Yes — EPR Plan filed

M/s E-Waste Recyclers India (HSPCB-registered recycler/dismantler)

E-Waste Management policy published on company website

Tata Electronics Pvt. Ltd.

Yes — via Tata Group E-Waste Management Policy

Follows Tata Group-mandated EPR registration and certified-recycler norms

Unlisted subsidiary; no separate public BRSR

Foxconn (Hon Hai) India operations

Yes — as Producer/Manufacturer under Schedule I

Engages CPCB/SPCB-authorised recyclers as part of global EHS policy

Disclosed via parent Hon Hai group sustainability reporting; limited India-specific public filings

5. OBSERVATIONS AND FINDINGS

• All five companies formally recognise EPR as applicable to their operations, consistent with the mandatory registration requirement under Rule 4 for Manufacturers and Producers of Schedule I equipment.

• Listed companies (Dixon, Syrma SGS) provide the most granular public disclosure, since BRSR filing is mandatory for SEBI's top-1000-by-market-cap listed entities.

• Companies working directly with named, state-authorised recyclers (Syrma SGS with Global Waste Solution; VVDN with E-Waste Recyclers India) demonstrate the clearest documentary trail required under the Rules — sales invoices, purchase proofs, and GST-linked recycling records.

• Unlisted or foreign-parented entities (Tata Electronics, Foxconn India) show compliance occurring at a policy level, but with less India-specific public data, reflecting a general gap that the CPCB's BRSR-linked disclosure push is intended to close.

6. CONCLUSION

The compliance landscape among leading Indian electronics manufacturers under the E-Waste (Management) Rules, 2022 shows near-universal formal recognition of EPR obligations, but uneven depth of public disclosure. SEBI-listed manufacturers with mandatory BRSR filing obligations, and those with named, state-authorised recycler partnerships, offer the most verifiable and documented compliance trail. Unlisted subsidiaries and foreign-parented operations, by contrast, tend to disclose compliance at a group-policy level rather than through company-specific, India-focused reporting. As BRSR-linked disclosure requirements extend further down the market-capitalisation ladder, this transparency gap between listed and unlisted manufacturers is likely to narrow.

This document summarises publicly available disclosures as of the search date and is intended for informational and academic use. It does not constitute verification of any company's live CPCB EPR registration status.

REFERENCES

1. Central Pollution Control Board (CPCB). E-Waste (Management) Rules, 2022 — Frequently Asked Questions, updated 23 January 2024.

2. Ministry of Environment, Forest and Climate Change, Government of India. E-Waste (Management) Rules, 2022, notified under the Environment (Protection) Act, 1986.

3. Dixon Technologies (India) Ltd. Business Responsibility and Sustainability Report (BRSR), FY 2023-24, filed with NSE, 3 September 2024.

4. Syrma SGS Technology Ltd. EPR Compliance disclosures, syrmasgs.com/epr-compliance; BRSR filings FY 2023-24 and FY 2025-26 (NSE).

5. VVDN Technologies. E-Waste Management Policy, vvdntech.com/e-waste-management.

6. Tata Group. E-Waste Management Policy Commitment Document, tatapower.com/who-we-are/company-documents.

7. Hon Hai Technology Group / Foxconn. Group sustainability and EHS disclosures (general industry reference).